

linePiece_260831a

size(1000, 1000);

background(255);

// draw 12 pairs of lines:

// the first part is a bold black line above the threshold

// the second part is a thin black line below

// we'll use some simple linear algebra to make this easier

// and a loop because i super don't want to do the math by hand

// output is randomized, so it comes out different every time

PVector thresholdStart = new PVector(width*0.25, height *0.20);

PVector thresholdEnd = new PVector(width*0.75, height*0.80);

int numLines = 12;

for(int i = 1; i <= numLines; i++){

PVector slope = new PVector(random(0.5, 0.9), random(-0.5, -1.5));

PVector center = PVector.lerp(thresholdStart, thresholdEnd, i/float(numLines+1));

float upperDistance = random(30, 200);

PVector lineUpper = PVector.add(center, PVector.mult(slope, upperDistance));

float lowerDistance = -random(100, 400);

PVector lineLower = PVector.add(center, PVector.mult(slope, lowerDistance));

stroke(#000000);

strokeWeight(10);

strokeCap(SQUARE);

line(lineUpper.x, lineUpper.y, center.x, center.y);

// lines become thin as they cross the red line

strokeWeight(3);

line(lineLower.x, lineLower.y, center.x, center.y);

}

// draw a thick red line down the middle, but a little askew

stroke(#ff0000);

strokeWeight(15);

strokeCap(ROUND);

line(thresholdStart.x, thresholdStart.y, thresholdEnd.x, thresholdEnd.y);

save("output.png");